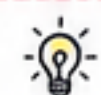




TECH WITH
SHIVANSH

LECTURE 12 : POWER QUERY – DATA IMPORT & CLEANING



Power Query

Get Data. Transform Data.
Combine Data. Load Data.
All without writing a
single formula!

1. WHAT IS POWER QUERY?

Power Query is an Excel technology that allows you to import data from multiple sources, clean it, transform it, reshape it and load it to Excel or Data Model.

2. WHY USE POWER QUERY?

- ✓ Import data from multiple sources quickly.
- ✓ Clean messy data in a few clicks.
- ✓ Remove errors, blanks, duplicates easily.
- ✓ Transform and reshape data as per need.
- ✓ Combine data from multiple files/sheets.
- ✓ Repeat the same steps on new data with Refresh.
- ✓ Saves time and reduces manual work.
- ✓ Powerful, consistent and reliable ETL tool.



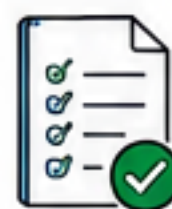
THE POWER QUERY PROCESS



Get Data



Transform
(Clean & Shape)

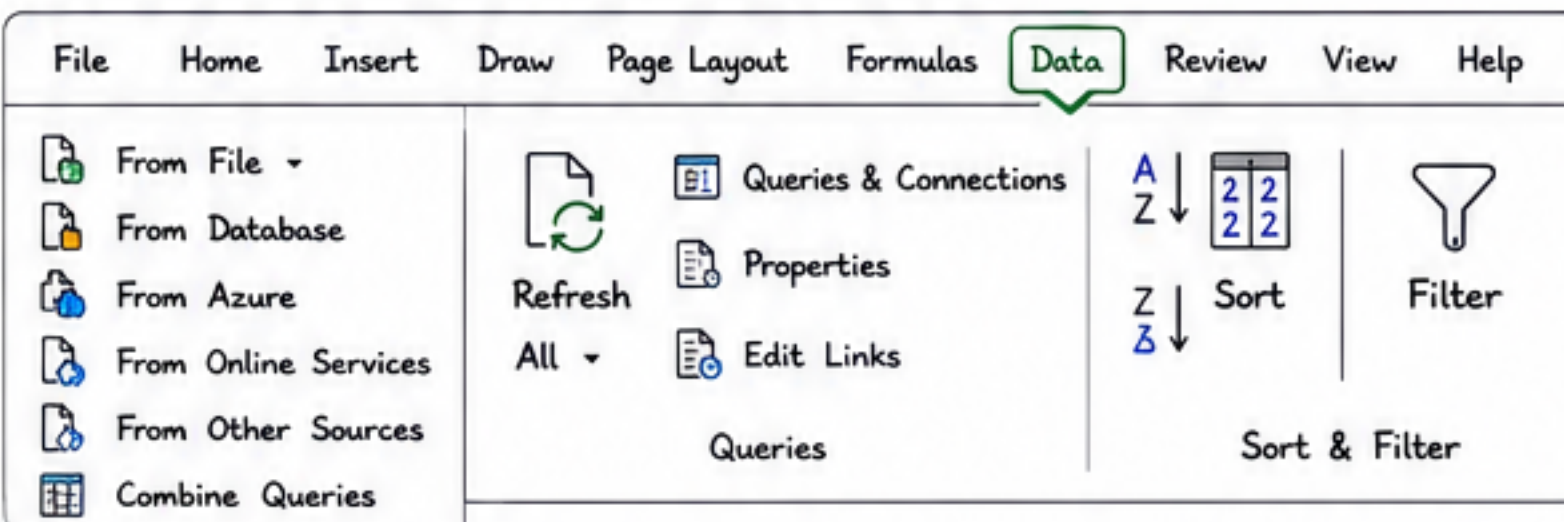


Apply Steps



Load &
Refresh

3. WHERE IS POWER QUERY IN EXCEL?



NOTE:

- Power Query Editor opens when you click on Get Data.
- Your steps are saved as a Query and can be refreshed anytime.

4. POWER QUERY EDITOR – OVERVIEW

Query Settings

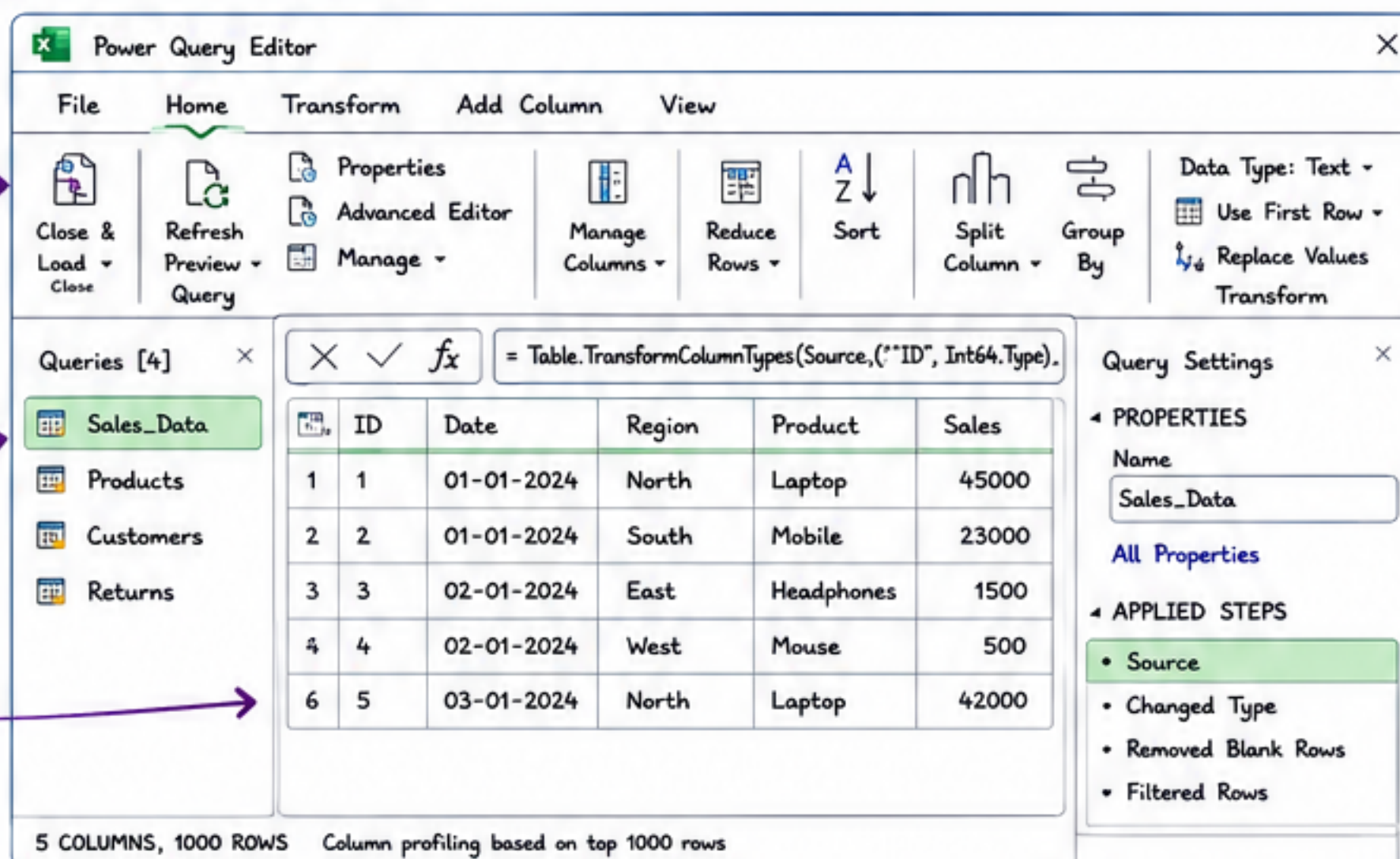
Manage your queries, rename, duplicate, delete, etc.

Queries Pane

All your queries appear here.

Data Preview

Shows a preview of the data and all applied steps.



Ribbon Tabs

Different tabs contain powerful tools for transformations.

Applied Steps

Shows all the steps you have applied in order.

Preview Pane

Preview of your data after each transformation.

5. GET DATA – DATA SOURCES

	From File	(Excel, CSV, Text, JSON, XML, PDF, Folder)
	From Database	(SQL Server, Access, Oracle, MySQL, etc.)
	From Azure	(Azure SQL, Blob, Data Lake, etc.)
	From Online Services	(SharePoint, Web, OData, etc.)
	From Other Sources	(ODBC, Active Directory, etc.)
	Combine Queries	(Append, Merge, Reference)

6. COMMON TRANSFORMATIONS (EXAMPLES)



Remove Rows

Remove blank rows or unwanted rows.

A	B
1	x
3	y

A	B
1	x
3	y



Remove Columns

Remove unnecessary columns.

A	B	C
1	x	p
2	y	q

A	C
1	p
2	q



Change Type

Change data type of columns.

A (Text)
123
456

A (Number)
123
456



Filter Rows

Filter data based on conditions.

A
10
30

A
20
30

(> 15)



Split Column

Split one column into multiple.

Name
Shivansh Yadav
Rahul Singh

First Name	Last Name
Shivansh	Yadav
Rahul	Singh



Merge Columns

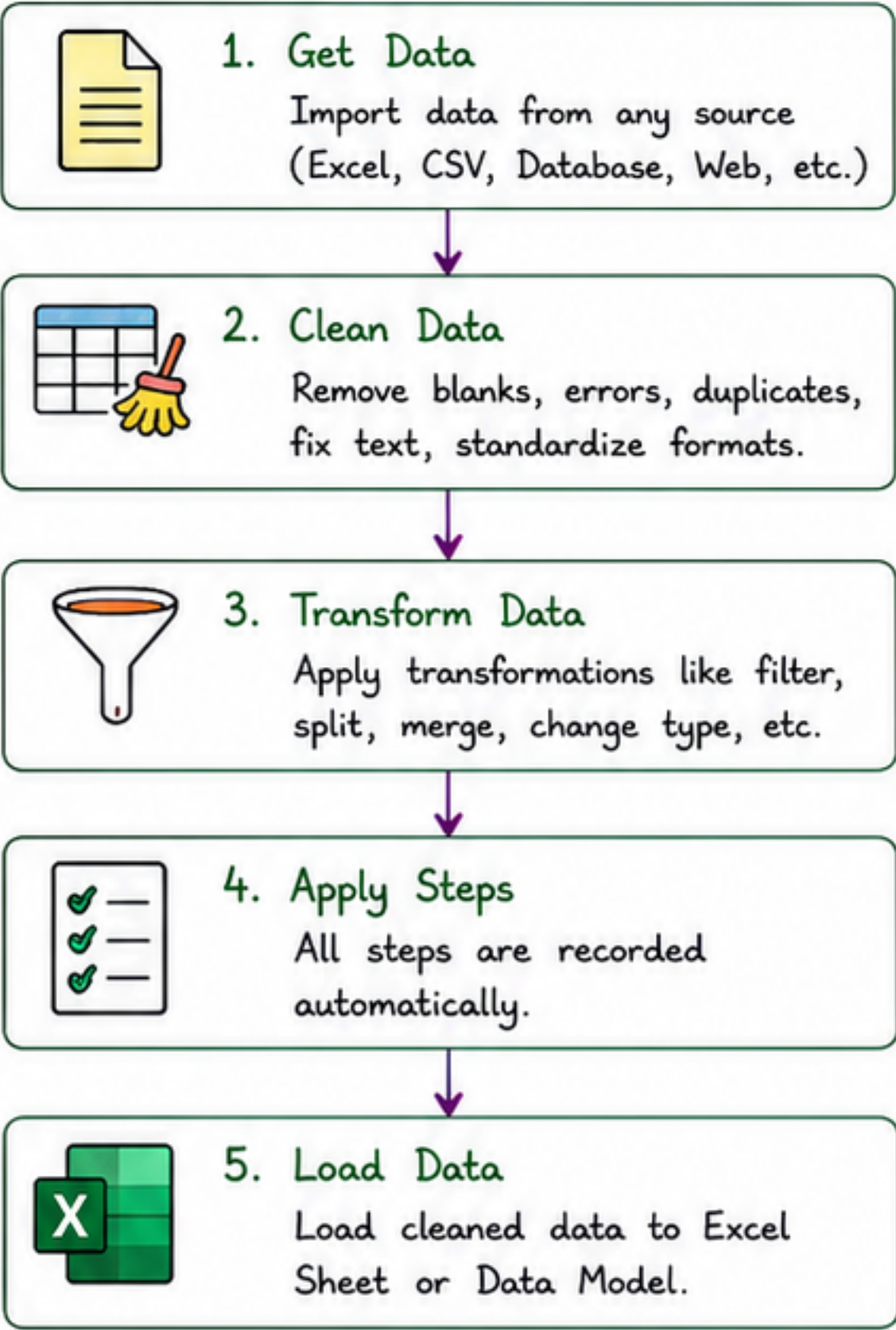
Combine multiple columns into one.

First	Last
Shivansh	Yadav
Rahul	Singh

Full Name
Shivansh Yadav
Rahul Singh



7. POWER QUERY WORKFLOW EXAMPLE



8. REAL LIFE USE CASES

- ★ Combine monthly sales files into one master file.
- ★ Clean customer data (Remove duplicates, fix text, standardize).
- ★ Prepare data for reports and dashboards.
- ★ Import data from databases and transform for analysis.
- ★ Automate repetitive data cleaning tasks.
- ★ Merge data from multiple departments / branches.
- ★ Create dynamic reports that refresh with new data.
- ★ Handle large datasets efficiently.

NOTE: Power Query helps you save time, improve accuracy and makes your data analysis process professional. 

9. INTERVIEW QUESTIONS

- Q. What is Power Query in Excel?
- Q. What are the benefits of using Power Query?
- Q. What are the different data sources you can import using Power Query?
- Q. What is the difference between Merge and Append in Power Query?
- Q. What is the use of Applied Steps?
- Q. How do you refresh data in Power Query?
- Q. What is the difference between Power Query and Power Pivot?
- Q. Can Power Query handle large datasets?
- Q. What happens when a source file is updated?
- Q. How do you remove duplicates in Power Query?



Think Like an Analyst.
Clean Data like a Pro.
Transform with Power!

10. POWER QUERY – KEY TABS & THEIR USAGE

Tab	Icon	Usage
Home		Manage queries, refresh, properties, choose data sources.
Transform		Perform all data transformations like filter, split, merge, change type.
Add Column		Add custom column, index column, conditional column, duplicate column.
View		Show/hide panes, column quality, column distribution, preview settings.
File		Close & load, options, manage settings.

KEY POINT:

- Each step you apply is recorded.
- You can modify or delete any step.
- All transformations are non-destructive.



11. POWER QUERY – IMPORTANT CONCEPTS

- ⇒ **Query** : A set of steps to get and transform data.
- ⇒ **Step** : A single action performed on data.
- ⇒ **Applied Steps** : The list of all steps performed in order.
- ⇒ **Data Preview** : Shows the result of applied steps.
- ⇒ **Refresh** : Re-runs the query and updates the data.
- ⇒ **Load** : Loads the data to Excel Sheet or Data Model.
- ⇒ **Close & Load To...** : Load to Sheet, Table or Data Model.

✓ STEP 1

✓ STEP 2

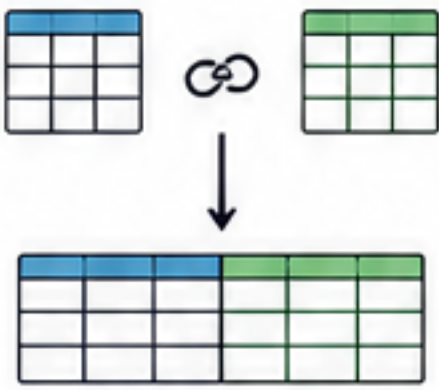
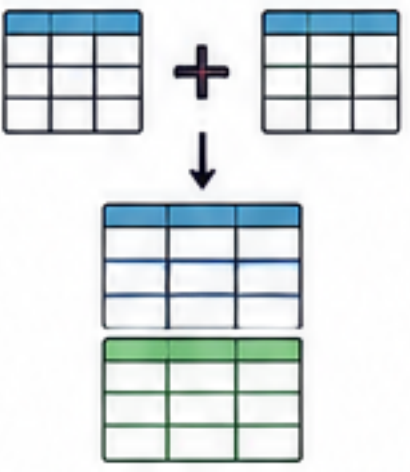
✓ STEP 3

✓ ...

✓ STEP N

APPLIED STEPS

12. MERGE vs APPEND

MERGE (JOIN)	APPEND (UNION)
- Combines columns from two tables based on a related column. - Similar to VLOOKUP. - Increases width (more columns). 	- Stacks rows from two or more tables with the same structure. - Similar to UNION ALL. - Increases height (more rows). 

13. POWER QUERY – ADVANCED FEATURES

- ★ **Custom Column** : Create new columns using logic or formulas.
- ★ **Conditional Column** : Add columns based on conditions.
- ★ **Group By** : Summarize data using aggregation (Sum, Count, Avg, etc.).
- ★ **Pivot Column** : Turn row values into column headers.
- ★ **Unpivot Column** : Turn columns into rows.
- ★ **Parameter** : Create dynamic values (e.g. file path, date, etc.).
- ★ **Manage Parameters** : Use parameters in queries.
- ★ **Combine Files** : Import multiple files from a folder and combine.
- ★ **Data Type Detection** : Automatically detect and set correct data types.

NOTE : Power Query is not just a cleaning tool; it is a complete ETL (Extract, Transform, Load) solution inside Excel.



14. POWER QUERY – DATA TYPES

Data Type	Icon	Usage / Description
Text	ABC	For names, cities, categories, codes.
Number	123	For amounts, quantities, values.
Date		For dates (e.g. Order Date).
Date/Time		For date and time values.
True/False	✓ ✗	For logical values.
Any	*	Detects and converts automatically.

15. POWER QUERY TIPS & BEST PRACTICES

✓ Always convert data types properly.

✓ Clean data as early as possible.

✓ Remove unnecessary columns and rows.

✓ Use meaningful step names.

✓ Disable "Enable Load" for intermediate queries.

✓ Use parameters for dynamic and reusable queries.

✓ Use Merge instead of VLOOKUP for better performance.

✓ Keep queries simple, consistent and well organized.

✓ Refresh data regularly to keep reports up to date.



Clean Data
Better Analysis
Better Decisions!

16. REAL TIME SCENARIO

You are a Data Analyst in an e-commerce company. You get multiple Excel files every month from different regions containing sales, returns and customer data.

Using Power Query you can:

- ✓ Combine all monthly files.
- ✓ Clean and standardize the data.
- ✓ Merge with master customer and product data.
- ✓ Create a single dataset for reporting.
- ✓ Refresh data in one click whenever new files arrive.



17. QUICK RECAP

- ➔ Power Query helps us get data from multiple sources.
- ➔ We can clean, transform and reshape data easily.
- ➔ All steps are recorded and can be reused.
- ➔ Queries are dynamic and can be refreshed anytime.
- ➔ It saves time and reduces manual work.
- ➔ It is a must-have skill for every Data Analyst.



Master Power Query Today,
Automate Your Data Work Forever!

18. INTERVIEW QUESTIONS – POWER QUERY

Q. No.	INTERVIEW QUESTION	EXPECTED ANSWER (Summary)
1.	What is Power Query in Excel?	➔ Power Query is an Excel tool used to import, clean, transform and load data from multiple sources.
2.	What are the benefits of using Power Query?	➔ It saves time, reduces manual work, handles large data, ensures consistency and makes data analysis easy.
3.	What are the different data sources in Power Query?	➔ From File, Database, Azure, Online Services, Other Sources (ODBC, Web, SharePoint, etc.) and Combine Queries.
4.	What is the difference between Power Query and Power Pivot?	➔ Power Query is used for data import & transformation. Power Pivot is used for data modeling and DAX calculations.
5.	What is a Query in Power Query?	➔ A Query is a set of steps applied on data to transform it as per our requirement.
6.	What are Applied Steps?	➔ Applied Steps show all the transformations done in order. They can be edited, deleted or rearranged.
7.	What is the difference between Merge and Append?	➔ Merge (Join) combines columns from two tables based on a common key. Append (Union) stacks rows from tables.
8.	What is the difference between Merge and VLOOKUP?	➔ Merge adds columns from another table (like a join). VLOOKUP looks up a value; Merge can bring multiple columns.
9.	What is the use of Refresh in Power Query?	➔ Refresh re-runs the query and updates the data in Excel with the latest changes from the source.
10.	What is the use of Close & Load To?	➔ Close & Load To allows you to load data to a specific location – Excel Sheet, Table or Data Model.
11.	What is a Parameter in Power Query?	➔ A Parameter is a named value used to make queries dynamic (e.g., file path, date, etc.).
12.	How do you remove duplicates in Power Query?	➔ Select the column(s) → Home Tab → Remove Rows Group → Remove Duplicates.
13.	What are the data types in Power Query?	➔ Text, Number, Date, Date/Time, True/False, Any, Currency, Decimal Number, Whole Number, etc.
14.	Can Power Query handle large datasets?	➔ Yes, Power Query is designed to handle large datasets efficiently with minimal memory usage.

19. COMMON ERRORS & SOLUTIONS

COMMON ERROR	SOLUTION
✗ Columns not recognized correctly.	✓ Change the data type from Data Type dropdown.
✗ Numbers stored as text.	✓ Use "Data Type" → "Change Type" → "Using Locale".
✗ Duplicate rows in data.	✓ Use "Remove Duplicates" from Home Tab.
✗ Extra blank rows/columns.	✓ Use "Remove Rows" or "Remove Columns".
✗ Query not refreshing.	✓ Check source connection and use Refresh or Refresh All.
✗ Merge giving unexpected results.	✓ Check join kind and data types of key columns.

21. KEY TAKEAWAYS

- ✔ Power Query is your go-to tool for Data Import & Cleaning.
- ✔ All steps are recorded and can be reused.
- ✔ It supports multiple data sources.
- ✔ It makes your data analysis faster, easier and reliable.
- ✔ It is an essential skill for every Data Analyst.

20. POWER QUERY SHORTCUTS

Alt + A + G	Open Get Data
Alt + H + R	Refresh Preview
Alt + H + Q	Manage Queries
Alt + H + C	Choose Columns
Ctrl + S	Save Query
F5	Refresh All



Learn Smart,
Use Power Query,
Work Less,
Achieve More!

22. PRACTICE EXERCISES (For You)

1. Import an Excel file and remove blank rows & columns.
2. Clean a messy dataset (remove duplicates, fix data types).
3. Merge two tables using a common key.
4. Append three sales files (Jan, Feb, Mar) into one table.
5. Create a parameter for file path and use it in a query.
6. Pivot a table using Power Query.